

B.Tech. Degree VI Semester Regular/Supplementary Examination May 2024

ME 19-205-0603 OPERATIONS MANAGEMENT
(2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

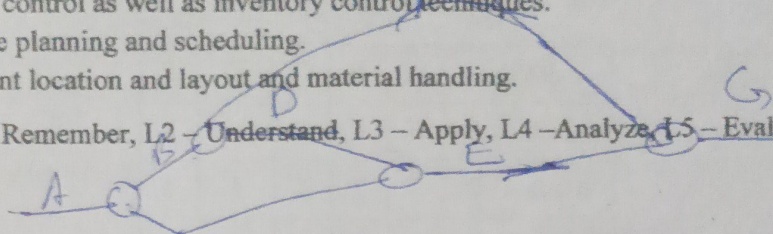
Course Outcome

On successful completion of the course, the students will be able to:

- CO1: Understand the different network techniques and forecasting methods.
- CO2: Learn production planning and control as well as inventory control techniques.
- CO3: Study the concepts of aggregate planning and scheduling.
- CO4: Get the basic knowledge of plant location and layout and material handling.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome



PART A (Answer ALL questions)

- (8 × 3 = 24)
- I. (a) What is the importance of slack/float in network analysis?
 - (b) What is Delphi technique?
 - (c) Briefly explain product life cycle.
 - (d) Explain Stores ledger with its format.
 - (e) What is the objective of Gantt chart.
 - (f) Define aggregate planning.
 - (g) How material handling equipments are classified?
 - (h) Explain sinking fund method to calculate depreciation.

Marks	BL	CO	PO
3	L2	1	1
3	L1	1	1
3	L2	2	1
3	L1	2	1
3	L1	3	1
3	L1	3	1
3	L2	4	1
3	L2	4	1

PART B

(4 × 12 = 48)

- II. (a) Differentiate between PERT and CPM.
- (b) Draw network and identify the critical path and find out the project duration.

ACTIVITY	A	B	C	D	E	F	G
DURATION	9	5	4	11	7	5	15
IMMEDIATE PREDECESSOR	-	A	A	B	C, D	B	F, E

OR

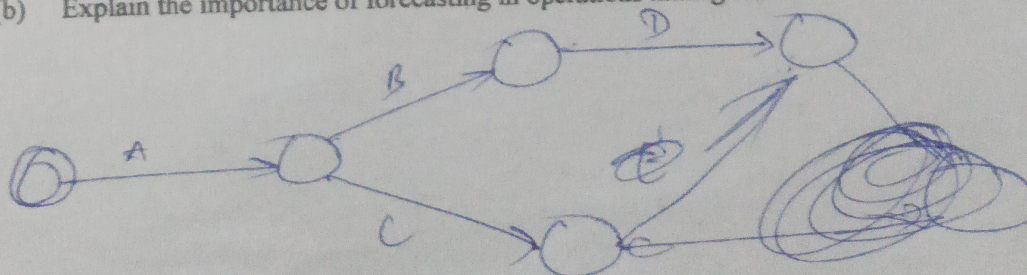
- III. (a) From the following data, predict the sales for the years 2008, 2009 and 2020 using linear regression method.

YEAR	2001	2002	2003	2004	2005	2006	2007
(Sales in '100 UNITS)	80	90	92	83	94	99	92

- (b) Explain the importance of forecasting in operations management.

8	L2	1	1,2
4	L2	1	1

(P.T.C.)



(b) What are the functions of production planning and control?

OR

- V. (a) Explain EOQ model of inventory control using a diagram.
 (b) An aircraft company uses rivets at an average consumption rate of 2500 kg per year. Each unit cost ₹30/kg and the company person estimate that its cost is ₹130 to place an order. The carrying cost of the item is 10 % of unit cost.
 (i) Find EOQ.
 (ii) How frequently the orders for the rivets be made.

VI.

JOBS	A	B	C	D	E
Machine 1	6	8	7	10	6
Machine 2	3	2	5	6	4
Machine 3	4	8	6	7	8

From the following data find the sequence, make span and idle time of machines using Johnson's algorithm.

OR

VII.

JOB Sequence	Processing Time (t)	Due Date (D)
A	2	7
B	8	16
C	4	4
D	10	17
E	5	15
F	12	18

From the following data find the optimum sequence of jobs using critical ratio method.

- VIII. (a) What are the objectives of plant layout?
 (b) Explain the different types of plant layout with block diagrams. Mention it's advantages and disadvantages.

OR

- IX. (a) Explain the factors involved in selection of a plant location.
 (b) Explain the different types of maintenance. Mention it's merits and demerits.

Blooms's Taxonomy Level's
 L1 - 30%, L2 - 40%, L3 - 30%.
